

TangiTeach: An Integrated Web and Hardware Platform for Empowering Instructors to Author Lessons Using Electronics-Based Tangibles

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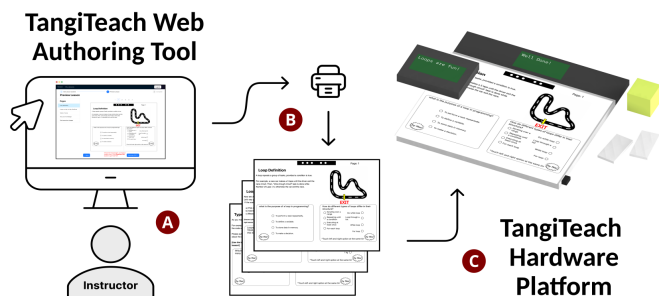


Fig. 1: TangiTeach enables interactive tangible lesson creation in minutes through physical structuring, embodied interaction, and reusable elements. (A) The web UI (Figures 5 and 2) supports template-based design without coding. (B) Printed pages with embedded codes connect to a reusable hardware platform (Figure 7). (C) The hardware automatically reads pages and supports tangible interactions (Figure 4A & Figure 6 in Appendix section), enabling rapid lesson authoring.

Abstract—Interactive tangibles have long shown promise for learning outcomes, yet few tools let instructors, not just researchers or developers, create their own tangible lessons. We present TangiTeach, an integrated end-to-end web and hardware platform that empowers instructors to author lessons with electronics-based tangible interactions. TangiTeach combines familiar web-based authoring metaphors (e.g., wizards, books) with reusable hardware that supports touch, gestures, and object recognition, without requiring coding or wiring. Through formative co-design sessions with instructors ($n=8$), we identified key challenges in tangible lesson authoring and refined the system to support spatial structuring of content, reuse and adaptation of physical components, and inspiration through community examples and AI-assisted ideation. In a subsequent evaluation ($n=16$), instructors created full interactive lessons, reporting strong motivation to further explore TangiTeach for explaining complex concepts to students. This work contributes empirical insights into how instructors approach tangible lesson design, and presents a proof-of-concept platform that enables the creation of interactive tangible lessons.

Index Terms—authoring tools, computing education, electronics-based tangibles, instructor-centered design

I. INTRODUCTION

Tangibles have been shown to provide playful, engaging, and memorable learning experiences for students [1]–[3], especially for exploring complex topics in computing education [4], [5]. For example, tangible metaphors like *Hanoi Towers*

can help students new to programming visualize recursion algorithms [6]. Tangibles integrated with electronics, can enhance learning [7] by engaging multiple senses and providing in-context feedback [1], [8], [9], as seen in interactive museum exhibits [10], [11] and children’s interactive books [12], [13].

Given the learning benefits of electronics-based tangibles, instructors are motivated to experiment with tangible learning [14], [15]. However, creating tangible lessons often requires significant time, technical expertise, and hardware skills [16], [17]. Although HCI and educational research has shown that tangible and embodied interactions can make learning engaging, most existing systems are designed by researchers or technical experts, leaving instructor-authored tangible resources underexplored [17]. As a result, few instructors are able to incorporate electronics-based tangibles into their teaching [2].

In this work, we adopt an instructor-focused approach to simplify the creation of interactive lessons with electronics-based tangibles. We designed an authoring tool that abstracts away hardware complexity, enabling instructors to focus on lesson design. Figures 5 and 2 illustrate the interface, while section III-A presents the process through a use case. Our core research question is: *How can we design an authoring environment that enables instructors with limited time and no prior electronics knowledge to create interactive, electronics-based tangible lessons?*

We present *TangiTeach*, an integrated web authoring interface and hardware platform for creating tangible interactive lessons. Our design draws on prior work in paper-based electronics (e.g., *TangiBooks* [18], *Paperpulse* [19]) that enable multimodal feedback. Unlike these systems, which emphasize technical potential, we center on instructors’ perspectives, developing a proof-of-concept authoring environment that supports end-to-end lesson creation. TangiTeach combines a web-based interface with a modular hardware platform (Figure 5) that removes the need for coding or electronics expertise. Using a notebook metaphor, it generates paper-based lessons that the platform can decode, enabling interactions with tangible objects (Figures 4A and 4B) and gestures.

After the first design iteration of TangiTeach, we conducted formative co-design sessions with 8 CS instructors and teaching staff to understand perceptions of tangible lesson design and refine the prototype. Participants found the authoring

system easy to use and engaging, but raised concerns about the time required to design lessons from scratch and the need for content inspiration. In response, we extended TangiTeach with community-based content sharing (e.g., curated examples) and Generative AI features for content inspiration.

Next, we conducted an observational study of lesson creation using TangiTeach with 8 instructor–student pairs (from K–12 and university settings). Despite limited electronics experience, all participants successfully created complete tangible lessons. Instructors were enthusiastic about TangiTeach’s tangible and embodied interaction capabilities to structure and explain complex concepts (e.g., Figure 4C), while the community and AI-supported features helped scaffold lesson creation and adaptation.

Our investigation shows that instructors without prior experience in tangibles or electronics can create interactive lessons using TangiTeach’s step-by-step structured yet flexible approach to tangible lesson creation. While tangible lesson authoring remains an emerging space, our findings suggest that AI-driven inspiration and community sharing can support instructors in exploring new teaching practices. Instructor feedback remains essential, and future work should further develop instructor-centered tangible environments. TangiTeach provides a foundation and design considerations for supporting instructors in this space.

This paper makes two contributions: (a) an end-to-end platform that enables instructors to author tangible lessons without prior electronics expertise; and (b) empirical insights into the practices and challenges involved in this process. Table II summarizes five recurring practices: physical element mapping, spatial layout authoring, interaction–behavior linking, reusable lesson scaffolds, and reversible configuration. By focusing on instructor-centered authoring, we frame lesson creation as a constructive process of building, adapting, and reusing representations of instructional content [20], [21].

II. RELATED WORK

This research builds upon prior work on the use of tangibles in learning settings and insights into creating electronics-based interactive tangibles lessons.

A. Tangible Interaction: Foundations and Learning Benefits

Education has long used physical activity and object manipulation to support learning [22]–[25]. Tangible and embodied interfaces help ground abstract concepts in physical action [26], Ullmer and Ishii [27] define Tangible User Interfaces (TUIs) as systems that couple physical representations with digital information through physical embodiment, manipulation, and sensor-based links between action and computation. Shaer and Hornecker [28] extend this view by describing tangibility as a spectrum that includes object manipulation, surface interaction, gestures, proxemics, and hybrid interfaces, as long as a stable physical-digital coupling is maintained.

These benefits are often explained through constructivism and embodied cognition. Constructivist theory holds that learners build understanding by integrating new experiences with

prior knowledge [20], [21]; tangibles support this through schema formation [29], shared representations within the Zone of Proximal Development [30], and scaffolding that can be gradually removed as learners gain competence [31], [32]. Embodied cognition further highlights how bodily action and sensorimotor engagement ground conceptual understanding [33], [34]. While these theories demonstrate why tangibles are pedagogically valuable, prior systems provide limited support for instructors to design or adapt such activities, motivating our focus on instructor-centered authoring in TangiTeach.

B. Use of Tangibles for Exploring Programming

Key ideas on tangible learning trace back to Froebel [35] and Montessori [36], whose manipulatives support abstract learning [37]. They have been applied in mathematics [38], programming [39], and data visualization [40]. Interactive systems extend these ideas by embedding sensing and feedback [41]. Early work such as the *Logo* turtle [42] introduced electronic manipulation for learning, followed by platforms including *AlgoBlock* [43], *Quetzal* [44], and *BlockyTalky* [45]. These systems make programming more accessible and engaging [46]. Recent work continues this direction, including *ML-Machine* [47] and *Tangible-MakeCode* [48], with comparisons highlighting the breadth of 18 tangible approaches [49].

Although prior work shows that tangible and embodied interaction can enhance exploration of programming, these systems are typically authored by researchers rather than instructors. Their interactions—such as block assemblies, tokens, or sensor-based artifacts—are often predefined and hard to adapt to different learning goals. Creating new lessons usually requires electronics expertise, hardware fabrication, or complex workflows, which are impractical for most instructors. Our work instead focuses on tools that support instructors to author themselves tangible activities aligned with their curricula.

C. Authoring Interactive Tangible Lessons: Challenges and Opportunities

Many hands-on computing platforms (e.g., *Lego Mindstorms* [50], *VEX* [51], *Arduino* [52]) require significant technical expertise, including wiring and coding. Craft-based systems such as *LilyPad* [53], *Electronic Poppables* [54], and *PaperPulse* [19] lower some barriers but still involve fabrication or deployment, and are not designed for instructional authoring. Similarly, *TangiBooks* [18] focuses on predefined interactions rather than instructor-created lessons. While prior work highlights tangible learning frameworks and assessment approaches [31], [55], [56], it offers limited support for instructor-led design. At the same time, instructors face time constraints [57], [58] while valuing ownership of materials [14], which current systems do not adequately support.

Our paper–electronics approach enables instructors to author customizable, embodied interactions (e.g., object placement, tokens, gestures). Unlike tablets or clickers, it supports multi-modal input and reconfigurable layouts, allowing meaning to emerge through physical interaction.

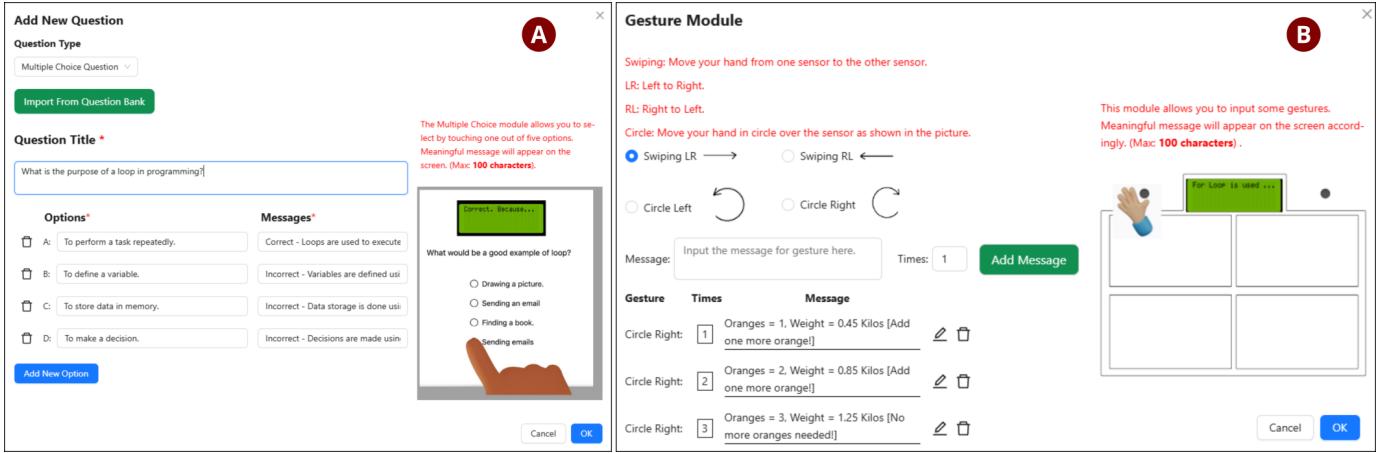


Fig. 2: TangiTeach web authoring UI for instructor-centered lesson creation. This image shows examples of interactive modules in TangiTeach. A) Matching questions created in the UI are printed and recognized by the hardware, providing immediate feedback. B) A gesture-based activity illustrates a *while* loop, where circular hand motions simulate adding oranges until a condition is met (incrementing *oranges* and updating *weight*). Instructors can make abstract programming concepts tangible without electronics expertise.

III. DESIGN AND IMPLEMENTATION OF TANGITEACH

To simplify instructor authoring of electronics-based tangible lessons, we considered three goals: 1) **Lower the technical barrier (DG1)** by removing the need for coding, wiring or complex setup; 2) **Support reuse with minimal configuration (DG2)** through a portable, low-cost platform using materials such as paper [18], [19], [54] and avoiding permanent assembly; and 3) **Enable customization (DG3)** by allowing instructors to tailor content and include self-check activities that have been shown to enhance learning [59], [60].

Based on these goals, we designed TangiTeach as a proof-of-concept system (Figure 1) with a web authoring tool that generates printable pages (Figure 5) and a reusable hardware platform (Figure 7) that connects wirelessly and supports touch, token detection, and lens-based interaction.

A. Example Interaction of Lesson Creation with TangiTeach

We next illustrate how instructors use TangiTeach to create a lesson. Consider Bob, an instructor with no electronics experience and limited time. He uses TangiTeach to create a lesson on *loops*. After logging in, he imports a pre-made lesson (Figure 5) and adds a page defining loops using a race car example (Figure 1C). He then adds an image and a matching question to assess understanding (Figure 2A), specifying prompts, options, and feedback.

Next, Bob adds a page explaining *for*, *for-each*, *while*, and *do-while* loops using a printed diagram with already-made tokens, and configures messages triggered by token detection (Figure 4A). He also includes a multiple-choice question with meaningful feedback.

Finally, he previews the lesson, reviews and prints the generated pages (Figure 5B; Figure 3), and tests them on the hardware platform (Figure 6C in Appendix section). Any updates in the authoring tool are automatically reflected on the device using WiFi connection.

B. TangiTeach Web-Based Authoring Interface

TangiTeach was designed with learning experts, designers, and HCI specialists to support instructor-centered tangible lesson authoring. The interface streamlines the process from creation to execution, foregrounding pedagogy while hiding electronics complexity.

1) **Lesson Design Process in TangiTeach:** A lesson is a set of pages organized around a shared learning goal. Each page supports static content (e.g., definitions, examples, images) and interactive elements, similar to tools like *Canvas LMS* and *SurveyMonkey* [61], [62]. These include customizable multiple-choice and matching questions. Inspired by *TangiBooks* [18], TangiTeach also supports object placement activities, where physical objects are recognized and feedback is provided (Figures 4A and 4B), enabling self-testing (DG3). (Section A details available modules.)

To support new users, TangiTeach tracks the evolving spatial layout usage and suggests missing content or interactions. During preview, it highlights empty regions and prompts additions, helping instructors create coherent and well-structured lessons.

2) **Metaphors for Guided, Reversible Tangible Configuration:** Creating tangible lessons can be challenging, especially for instructors without prior experience. To lower this barrier, TangiTeach uses a wizard-style interface that applies key design principles, such as providing user control and flexibility. The step-by-step guidance helps instructors to move between steps and revise decisions as needed (DG1), with each screen presenting focused actions to support deliberate choices.

TangiTeach also adopts the metaphor of a physical workbook with removable pages (DG1). Each page is divided into four sections (Figure 7A; Figure 5B), mirrored by the hardware configuration. This structure simplifies use while remaining customizable through the modular design.

Interactions occur on printed pages, with hardware modules reused across lessons (DG2) and electronics hidden from users. The system assigns a code to each page, enabling preview and printing on standard printers (Figure 5B).

Board Setup Instructions

Lesson Title: Introduction to Loops

Lesson Description: This lesson explains the fundamental concepts of loops.

Pages: 4

Interactive Components: 9

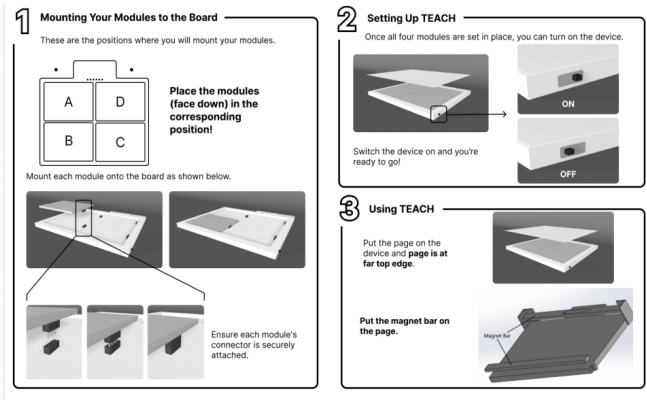


Fig. 3: Each lesson produces step-by-step IKEA-style set up instructions showing hardware module placement (see Figure 7A and 7F in appendix), lesson details, and page codes. Users can simply place the lesson page on the device, secure it with the magnet bar, and the system fetches the customized lesson information to the device, without any configuration or wiring.

3) **Customizable Interaction–Behavior Linking and Layout Design:** While simplifying the range and complexity of tangible objects was key to ensuring ease of use, TangiTeach still offers flexibility and control. Users can create lessons from scratch or import from a curated lesson bank, and can add sketches or additional materials directly to the pages without disrupting lesson flow. The authoring interface follows a step-by-step process to create content and map physical interactions, allowing users to move forward or backward to make corrections. A WYSIWYG preview feature is included to review lessons before printing.

TangiTeach allows users to customize every interaction with the device and lesson. The multiple-choice module (Figure 2A) also supports yes/no questions, while the object placement module enables the tagging of multiple physical artifacts with RFID, supporting flexible physical–digital coupling and reconfigurable interaction–behavior mappings (e.g., cards, or cubes as shown in Figure 4B and 6B in Appendix section). The tool provides guidance on how to use these RFID tags and customize the displayed message when placed over the sensor.

4) **Implementation of the Web Authoring Interface:** The frontend (using *React*) handles rendering and user input, while the backend (using *Express*) manages API requests and a MySQL database. The hardware communicates via MQTT: upon page detection, it sends a request, an AWS script retrieves interaction data from the database, and returns it to the device.

C. TangiTeach Hardware Platform

The TangiTeach hardware platform is a custom device integrated with the web authoring tool (see appendix A for details about the modular and expandable design of this device).

1) **Key Features and Setup:** The device uses a foam-board base with a holder for letter-size pages (Figure 7A). When a

page is placed, it is detected via printed markers and lesson data is retrieved over Wi-Fi, without any buttons or menus.

Users select options (e.g., quiz answers), and feedback is provided via a 128 × 64 OLED screen and buzzer (Figure 7D–E). The system runs on an ESP32 (C++/Arduino), powered by a 1000 mAh battery, and supports four modular slots.

Pages are secured magnetically with two foam bars, with modules placed beneath. Page detection uses seven light-dependent resistors sensing printed patterns. The current implementation supports up to 64 pages per lesson.

In our first iteration, we developed three interchangeable modules, each of which supports configurable actions through the web-based authoring UI:

- **Multiple Choice Question Module:** detects touches on paper and displays feedback for the selected choice on the TangiTeach screen (Figure 7D, appendix).
- **Matching Question Module:** supports simultaneous touches on two columns on paper to detect a match between a question and the answer.
- **Object Placement Module:** detects RFID-tagged objects such as cards (Figure 4A) or cubes (Figure 4B) for kinesthetic interaction.

Additionally, we include modules to provide enhanced kinesthetic interactivity:

- **Hand Gestures:** detects gestures via top-left and top-right corner sensors (Figure 7C, appendix). Users assign actions (e.g., hand swipes or circles) in the interface (Figure 2B), with responses varying by repetition (e.g., increasing values in a loop).
- **Tangible Lens Device:** A small, battery-powered screen device displays instructor-defined messages based on placement on the page (Figure 4C), enabling contextual explanations. For example, in a lesson on conditionals, positioning the device over different examples can reveal additional explanations for each one.

TangiTeach has low technical requirements, needing only a web browser with internet access. It uses inexpensive, off-the-shelf components (e.g., standard paper, no special inks or custom PCBs), making it cost-effective compared to tablets or mobile devices. The hardware is reusable across lessons, requiring only changes to printed pages (DG2).

IV. FORMATIVE CO-DESIGN SESSIONS FOR LESSON CONTENT CREATION WITH TANGITEACH

After the first design iteration, we conducted co-design sessions to explore how instructors create tangible lessons with TangiTeach and identify improvements.

A. Participants and Procedure

We recruited 8 participants (4 instructors, 4 teaching assistants or TAs; 6M, 2F, aged 26–60) via university mailing lists and personal contacts. The study was IRB-approved and all participants provided informed consent. Instructors had experience in programming education, with limited exposure to tangible tools. The TAs were postgraduate CS students with

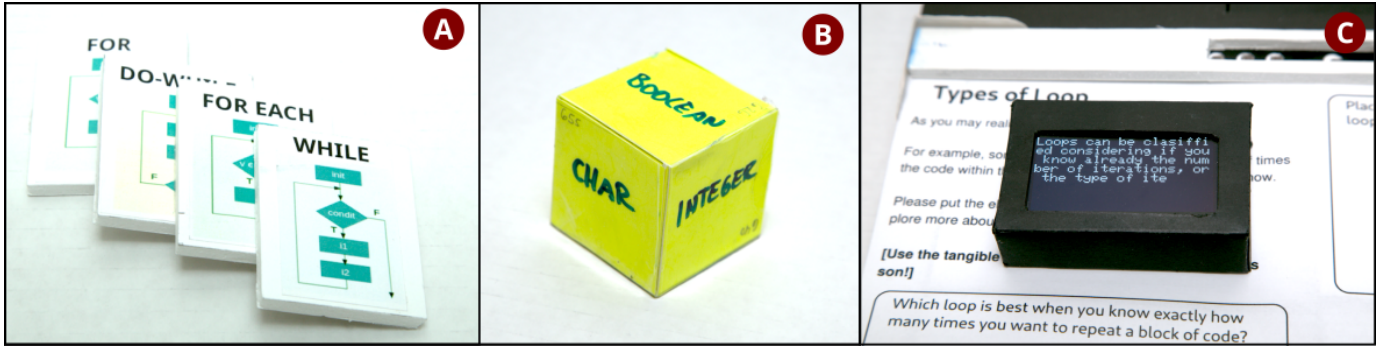


Fig. 4: TangiTeach interactive elements. A) RFID-tagged tokens (*TangiTokens*) display information when placed on the object placement module. B) A cube module with RFID tags on each face retrieves different data. C) A tangible lens displays configurable messages when placed over RFID tags. These modules enable hands-on, tactile interaction with printed lessons.

2–3 years of teaching experience leading programming tutorials, preparing instructional materials, planning lessons, and grading assignments, making them appropriate participants for evaluating an instructor-centered authoring workflow.

We conducted four exploratory co-design sessions with 8 participants, working in pairs of one experienced instructor and one TA, to study early-stage lesson design. Drawing on co-design approaches [63]–[67], we invited participants to discuss teaching needs and suggest improvements to TangiTeach’s usability. Our goal was to examine perspectives on the authoring workflow and hardware platform, and identify challenges and opportunities for refinement. Sessions and user interactions with the device were recorded to facilitate analysis.

Each session began with a demographic and experience questionnaire, followed by two 10-minute tasks to familiarize participants with the system. They first customized a lesson on loops by adding or rearranging a page, writing an explanation, and creating a multiple-choice question (Figure 2A). Second, they modified a variables lesson by replacing a static explanation with an RFID cube-based interactive setup (Figure 4B). After each task, participants printed and tested their lessons on the hardware platform using the setup instructions (Figure 3). Setup was minimal: they placed modules as instructed and turned on the device; Wi-Fi had been preconfigured. At the end, we conducted interviews about challenges and opportunities to improve authoring. Questionnaire data were analyzed using descriptive statistics, and interviews were transcribed and analyzed using an inductive approach [68].

B. Key findings

Participants viewed TangiTeach as a promising way to create tangible lessons without needing to code or wire hardware. As one noted, “*tangibles are something I imagine using at some point in my class. But, without that [TangiTeach authoring tool], there is zero chance I will use them!*” (Pair #4). For most, this was their first time authoring a tangible activity, and the sessions surfaced a consistent need for support during lesson creation. Participants emphasized that starting from a “blank slate” felt daunting and expressed a strong preference for examples, templates, or prompts to help them conceptualize tangible interactions and structure lesson content.

Based on their current lesson-planning practices, participants highlighted two opportunities for further enhancing and streamlining the TangiTeach authoring workflow.

1) Leveraging Other Instructors for Lesson Content::

All participants agreed that creating new lesson content is time-consuming and may discourage the use of tangibles. Pair #2 noted, “*If I had to create something [new], I need to think first about how much time it will take.*” Pair #4 echoed this sentiment, saying, “*[I don’t use tangibles] because I’m lazy... creating tangible lessons is harder than making a simple PowerPoint.*” This challenge was evident during Task 1, which required adding, rearranging, and explaining a loop concept using TangiTeach. Most time was spent discussing lesson ideas rather than implementing them in the authoring UI.

Considering these challenges, participants suggested leveraging examples from other instructors to support lesson creation: “*I feel that being able to have a library that allows you constructively create lessons centered around play is interesting and hard enough.. I don’t think like a have good things in my past experience like that [using tangibles].. the problem is I don’t know how!*” (Pair #4). Access to others’ experiences would help them understand possibilities: “*my first thought is I’ve seen people be more creative with this kind of thing [tangibles]... after seeing some of those you start really see [how] could be really really good!*” (Pair #3).

Pair #3 proposed a library of reusable examples: “*If you had something like a library people can use or just tweak, it’d be great.*” However, such a library should include ways to assess credibility and reputation: “*I’d personally feel disinclined to use other people’s lesson plans unless they come from people that I know. If someone said.. Oh, it is UPENN; I would say hey, can I have your stuff?... compared to a lesson from an anonymous thing from a Reddit editor or a wiki... colleagues, definitely will do!*” (Pair #4).

2) Seeking Inspiration for Tangible Lessons from Generative AI: Participants also suggested integrating Generative AI tools for content inspiration, particularly for creating self-testing questions. Pair #4 noted: “*I appreciate the idea of being able to use the large language models to do my work. If you can get it so that it is in a specific format where I can just click import. I would actually consider using that over actual*

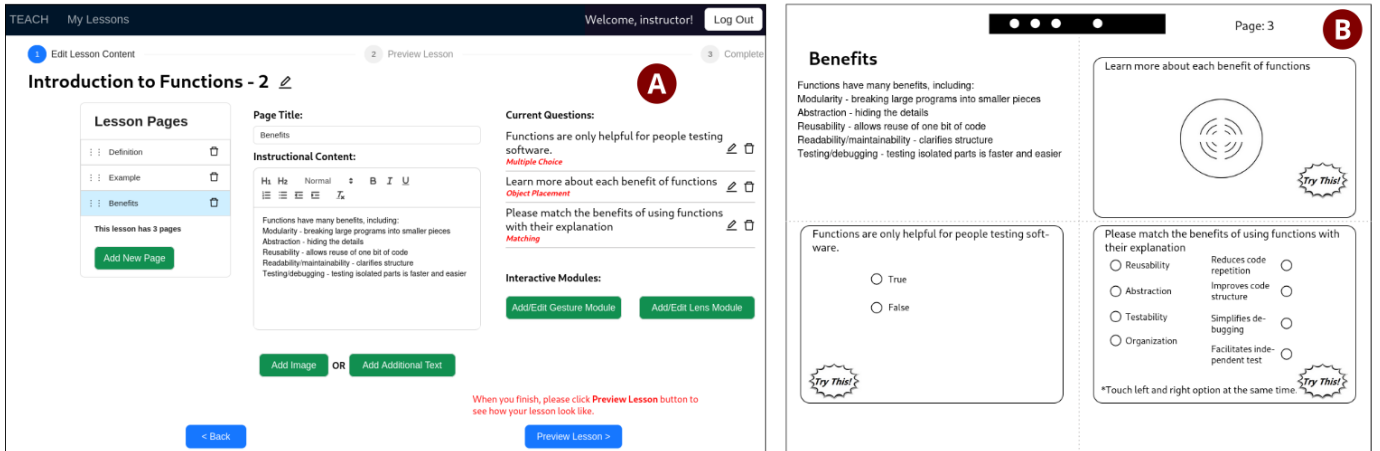


Fig. 5: Example of an instructor authoring an end-to-end lesson with TangiTeach. G8 created a lesson about functions in programming. A) The GUI page (*Benefits*) includes yes/no, matching, and object placement with *Tangible Tokens*; B) This GUI generates a page that is automatically exported as a PDF with an optically-encoded page number, while messages are transmitted to the device via the cloud.

PowerPoint... In this, I can just say... Hey, AI, do my job!". Pair #1 echoed this, highlighting AI's role in generating and adapting resources: *"You could even try importing something that already exists and then throw that into the AI and customize it or generate things that are similar back as well... take some of the things from the library. Throw them into the AI and say, generate 20 questions like this"*. Participants viewed AI as a way to explore examples, spark creativity, and generate content. Thus, we expanded our implementation to include generative AI in the next iteration.

C. Extending TangiTeach with Inspiration for Lesson Content

Based on participant feedback, we extended TangiTeach with two features to support lesson creation: a community lesson bank and an AI-powered content generator. The lesson bank enables instructors to browse, share, and evaluate lessons, including author details, ratings, comments, and filters, as well as a *question bank* for importing individual questions without full lessons, supporting resource reuse and community building [69], [70]. To support content inspiration, we integrated a simple AI module (powered by Google's Gemini API) that generates self-assessment questions from prompts, providing up to five answer options with explanations and one-click import. Instructors can refine or regenerate content iteratively, drawing on prior work in AI-supported teaching [71]–[74].

V. OBSERVATIONAL USER STUDY

We conducted an observational user study to evaluate how instructors without formal training in electronics used TangiTeach. We also considered perspectives from TAs and students who interacted with instructor-created lessons and observed their engagement. This dual perspective provided insights into both instructor usability and content creation, as well as the perceived opportunities for classroom use.

A. Participants

We recruited 16 participants (6M, 10F) from instructor and student lists, personal contacts, and local high schools. As shown in Table I, participants included 5 university lecturers

(2–16+ years of programming teaching experience, one with K–12 experience), 6 post-graduate TAs, 2 K–12 teachers, and 3 undergraduates with some programming exposure. None had participated in the formative co-design sessions. This study was approved by our IRB and all participants provided informed consent. Following our earlier procedure, participants worked in instructor and student/TA pairs (Groups #1–#8 from now G1 to G8) to foster discussion on authoring tangible lessons and their perceived learning impact.

B. Procedure

The study examined the authoring and use of tangible lesson materials rather than their effects on student learning. Although participants in the instructor category held different formal roles, all had teaching experience, such as leading tutorials, preparing instructional materials, planning lessons, or grading assignments. We therefore refer to them collectively as instructors in the remainder of the paper.

Each pair completed authoring tasks using TangiTeach hardware, tangible tokens, a web authoring tool, and a printer. Setup was minimal, requiring participants to position modules and power the system, as Wi-Fi was pre-configured. To examine how instructors approached lesson creation with an unfamiliar tangible modality, we used an observational protocol, with each session averaging about 1 hour. We collected data via questionnaires (e.g., demographics, teaching experience, and familiarity with electronics or tangibles), screen recordings, think-aloud verbalizations [75], [76], and follow-up interviews. Participants considered two personas: (1) Bob, a student who struggled with traditional programming learning; and (2) Alice, a time-constrained instructor with limited electronics experience. Semi-structured interviews explored prior experience and perceptions of TangiTeach as a tangible authoring platform, including its web UI, hardware, usefulness, and limitations.

C. Data Analysis

Consistent with our first study, we conducted an inductive qualitative analysis [68]. The first author transcribed all

Group	Role	Age	Gender	Experience
G1	Instructor	31–35	M	2–3 yrs exp, Python prog
	Student	22–25	M	UG CS student
G2	Instructor	22–25	W	2–3 yrs exp, basic C++ prog
	Student	22–25	W	Grad stats student
G3	Instructor	50+	M	16+ yrs exp, basic prog (univ/K–12)
	Student	20–21	W	UG math student
G4	Instructor	22–25	M	2–3 yrs exp, prog
	Student	22–25	M	MSc CS
G5	Instructor	26–30	M	2–3 yrs exp, web dev & SW eng
	Student	22–25	W	UG chem student
G6	Instructor	36–45	W	11–15 yrs exp, SW eng
	Student	31–35	W	PhD chem student
G7	Instructor	26–30	W	2–3 yrs exp, MSc CS
	Student	20–21	W	UG CS student
G8	Instructor	31–35	W	Sec. school math/sci teacher, 4–6 yrs exp
	Student	36–45	W	Elem. teacher, 16+ yrs exp

TABLE 1: Our user study had pairs of participants (instructor/ student) who explored TangiTeach and its authoring features together. All instructors were new to creating electronics-based interactive lessons.

sessions and began open coding to identify initial themes, using an evolving codebook refined throughout the process. Codes were iteratively compared, merged, and differentiated using constant-comparison techniques, and segments were color-coded to cluster emerging patterns, contradictions, and outliers. We then applied axial coding [68] to group related codes into higher-level categories while retaining links to the raw data. The research team met regularly to discuss interpretations, resolve discrepancies, and refine thematic structures. To strengthen the validity of emerging themes, video and screen recordings were used to triangulate participants’ reported perceptions with observed behaviors.

D. Usability Tasks

We designed TangiTeach to enable instructors to create end-to-end tangible programming lessons without coding. To evaluate this, participants completed three tasks to explore the authoring UI and individual lessons. These tasks, informed by pilot studies, aimed to assess usability, mental models, and user perceptions. A complete description of all tasks is provided in the appendix section B.

Task 1: The student participant led this task to explore a pre-made lesson to become familiar with the platform (5 minutes max). They interacted with the hardware, modules, and tangible tokens using printed pages on loops, including definitions and loop types (for, while, for-each, do-while) (Figure 4B).

Task 2: The instructor led this task using the web-based UI to modify a lesson about loops by editing feedback in a self-testing question and observed how changes appeared on the hardware (5 minutes max).

Task 3: Participants together created a full lesson on *functions*, adding pages and questions while using resources such as a lesson bank, AI features, and provided materials to support content creation (30 minutes max).

Participants used multiple resources to create lessons, including Google, built-in AI, a question bank, and provided

materials with examples and prompts. These were offered to support inspiration, as the focus was on evaluating the authoring process rather than content quality.

E. Findings

Based on task observations and interviews, we synthesized findings about instructors’ prior attitudes toward tangibles, their initial experiences authoring tangible lessons using the enhanced version of TangiTeach, and how tangibility shaped their perceptions of feasibility and pedagogy.

1) Initial Reactions to tangible lessons: The majority of participants (13/16) reported no experience with electronics at all, but all of them showed interest in using electronics-based tangibles for learning (8/16 were very or extremely interested). Still, many pointed to past negative experiences or a lack of prior exposure as reasons for hesitation. As the instructor in G6 noted: *“the availability is one problem for not using more tangibles in learning. So, like when we took courses [as students], we didn’t have these things. So I think that’s the first thing you fall back on, your own experiences.”* Another instructor (G5) reflected on the effort required to design such lessons, remarking that it, *“would take me the whole semester just to even think about how to do it.”*

By exploring TangiTeach, participants appreciated the refreshing learning approach tangibles could provide. Some were surprised: *“this is the first time I’m coming across any kind of tangible device... I never thought that [teaching a] programming language which requires software and computers could be taught by using tangibles!”* (G2); *“I was surprised to realize that programming can be taught with tangibles”* (G7). Others appreciated that TangiTeach included the electronics ready to use: *“because those things require you to know how to design workable tangible devices. Probably the really hard thing for us, especially [that] I don’t have too much knowledge about electronics.”* (G4). A student echoed the “joy” of tangibility: *“for me I learn by manipulating and touching.”* (G3). A K–12 teacher was eager to adopt TangiTeach right away: *“I mean, the big question mark for me is, where do I get one?”* (G8).

2) First-time authors successfully created tangible lessons: Participants initially expressed surprise at how feasible it was to author tangible lessons using TangiTeach, describing the platform as *“very easy to use”* (G8) and expressing interest in trying it independently. Others highlighted its modularity, that supported reuse across lessons: *“[the] modular thing [in TangiTeach] is very nice and it can be customized”* (G1). After completing the authoring tasks, participants shared their first impressions of TangiTeach, expressing surprise at the feasibility of creating tangible lessons without technical expertise and enthusiasm for the pedagogical opportunities it enabled. G7 discussed and designed a fully-fledged lesson on the importance of positional arguments for functions in programming. The lesson featured instructional content sourced from Google and two interactive tangible activities: a matching question that connected several different outputs depending on the position of variables. For example,

different outputs for `function(a, b, c)` compared to `function(a, c, b)`.

The K-12 secondary and primary instructors (G8) noted that TangiTeach, *“paired with the platform that provides a focused experience, is better than an iPad in which [high school] students randomly google something... Or play a dumb video or take pictures.”* They favored tangibles because *“physicality triggers memory, but it also triggers meaning.”* One teacher gave an example of how they will use the physical element mappings and reusable lesson scaffolds afforded by tangibles: *“In math, sometimes it’s difficult to understand if you have to compare and say which is bigger?... It could be difficult for students to understand... with [physical] tokens and with a circle or cake, you see what is bigger, what is smaller, and then they can select this is bigger.”*

3) User Behavior and Perceptions of Tangible Authoring Workflows: We observed that participants’ actions prompted reflection as they engaged in their first tangible authoring experience. We highlight key findings showing how TangiTeach supported participants in exploring interaction mappings and fostered a sense of empowerment as they engaged in the creation of tangible lessons.

Assign or redefine meanings for tokens, pages, or modules. Participants valued the ability to assign and redefine semantic meaning, treating tangible elements as reconfigurable physical-digital couplings during authoring. For example, G7 discussed and created their own tokens by writing relevant terms (e.g., Function, Procedure, Arguments, and Result) on blank tokens, which the platform then interpreted and explained by displaying the corresponding definition when each token was placed on its surface. The flexibility offered by TangiTeach aligned with participants’ expressed desire for customization, as one instructor noted, *“I’m looking forward to seeing how I can customize RFID to other objects”* (G1). Participants treated tangible elements as mutable representations and controls, reassigning their semantic roles to align with evolving pedagogical intentions.

Arrange content and interactions across physical pages and regions. Participants used physical pages as spatial organizers, leveraging layout, grouping, and ordering to structure explanations, compare concepts, and externalize instructional intent. Rather than static displays, pages functioned as manipulable cognitive scaffolds and meaningful regions within a lesson. As one participant noted, *“You could arrange things spatially, like when you sketch on paper before coding”* (G6). Another described pages as structured canvases, indicating *“I would suggest structuring [pages] with sections, maybe theory at the top and questions below. You could add more modules to include more information on the same page”* (G1). Across groups, explanations, matching activities, and vocabulary exercises were distributed across pages, supporting reuse and reconfiguration and reflecting participants’ appreciation of the platform’s modularity as *“very nice and customizable”* (G1).

Map physical actions (placement, touch, gesture) to system responses. Participants demonstrated that they understood and could leverage mappings between physical actions and

system responses. For example, one group repurposed tokens originally designed for loops (Figure 7A) to create novel activities that extended loop explanations and adapted them to illustrate function-related concepts. Participants also explicitly connected physical interaction to meaning-making: *“For me, I learn by manipulating and touching”* (G3), while K-12 instructors emphasized that *“physicality also triggers memory, but it also triggers meaning”* (G8). These comments suggest that the relationship between physical actions (e.g., placement and manipulation) and digital responses was intelligible and meaningful to participants.

Start from example lessons or templates and adapt them. Most groups (6/8) explored the shared lesson library and valued its potential to save time and provide inspiration. Participants described access to example lessons as efficient and supportive, with one noting, *access to a shared library is a very good thing... [it] can enhance the learning ability without consuming time”* (G1). They found importing lessons straightforward: *“it was easy for me, in short”*, G2. Participants also appreciated being able to focus on lesson content rather than technical setup (*“I’m focusing more on the lesson content... the part of creating a tangible is something which is handled by the app on its own”*, G5). K-12 instructors similarly valued the ability to reuse trusted materials instead of *“reinventing the wheel”* (G8). Participants frequently treated these example lessons and templates as starting points, adapting them to their own instructional contexts. Several compared this process to reusing familiar presentation templates, with one instructor explaining, *“It’s like finding PowerPoint templates when you have no idea for a fun presentation... the thing you needed was just that initial project made by someone else to help you get started creative wise. So I can see [TangiTeach] in the similar direction”* (G6). Together, these accounts suggest that participants used example lessons as reusable scaffolds, enabling iterative adjustments to physical form and associated meanings, rather than treating tangible designs as fixed artifacts that are typically revised with difficulty once established.

Reconfigure physical elements during authoring without permanent commitment. Participants valued the ability to iteratively reconfigure physical elements during authoring without committing to a fixed configuration. Modularity was repeatedly highlighted as enabling this flexibility, with participants appreciate that the system was “customizable” and “easy to revise.” Several groups refined lessons through successive edits, repurposed tokens, or adjusted content with AI assistance before finalizing designs. One instructor emphasized the benefit of abstracting technical complexity, stating, *“I’m focusing more on the lesson content... the part of creating a tangible is something which is handled by the app on its own”* (G5). This iterative behavior suggests that TangiTeach could support low-risk experimentation with physical elements and interactions. All of these findings are summarized in Table II.

4) Perceptions of Long-Term Use of TangiTeach: Participants saw TangiTeach as lowering the practical cost of tangible interaction, supporting integration into existing teaching workflows. AI-based suggestions provided lightweight guidance,

while customization was viewed as key to aligning tangibles with their teaching preferences and practice.

Appropriating TangiTeach in Practice. Participants viewed TangiTeach as a time-saving enabler that made tangible lesson design approachable: *“this one [TangiTeach] is far above my expectations... a great example [of] how to teach with tangible things”* (G4). They appreciated that lesson editing and hardware setup were automated: *“If you’re an instructor, you don’t want to deal with the physical issues”* (G6). The interface metaphors also resonated with participants: *“The wizard is very nice, and this step-by-step thing, I like that”* (G1). The book-like printed pages were perceived to be intuitive and familiar. Others were curious about new metaphors like the Tangible Lens and were curious about exploring other design options with them (G1).

Although some participants noted scalability concerns for tangibles in large classrooms, echoing prior work [26], [77], most of them expressed enthusiasm for TangiTeach as a *“new way to not only interact with the lesson, but also have fun with it as well”* (G5). G3 highlighted the appeal of TangiTeach for use in smaller settings: *“[TangiTeach] is interesting... it’s fun to do something at the table... roll dice or play games, flip coins”*. Others envisioned uses beyond programming: *“I see the potential of introducing these tools [for] other things, like even board games. Because you can design your own game on that system”* (G1).

Instructors valued content inspiration from AI.

As instructors are getting more familiar with GenAI tools in their teaching [78], [79], we found that most were enthusiastic about trying the AI feature: *“Most of the time AI can come up with very creative ideas or... questions or lessons in much less time”* (G1), even though some adjustment was needed *“[AI] gives interesting examples”* (G3). Participants expressed that AI could even further lower the barrier to authoring tangible lessons: *“the AI generated feature is really powerful... can probably add more AI empowered features on this not only for the questions but also help us to refine the [other] content”* (G4). Another participant proposed: *“If I can import [my lesson] to the system and it can give me ideas like, hey, you could do it this way...I gave you my lesson for tomorrow and you give me some ideas of how I can make it tangible”* (G5).

Instructors want control over customization and interaction design of lessons. Participants expressed a strong desire for lesson customization. Given an open-ended prompt on functions with an intentionally undefined scope, participants created detailed, engaging lessons, creatively leveraging TangiTeach’s interactive features and adapting their pedagogical knowledge to this medium. For instance, over half of the groups (5/8) started lessons from scratch, refining them with AI assistance and subsequent personal edits. The tangible tokens were highly popular. Participants mentioned personalization of content as a driver for design: *“so I’m looking forward to seeing how I can customize RFID to other objects”* (G1). Instructors also wanted to customize interactions: *“as we are working with different kinds of courses... maybe there are differences in the interactions”* (G2).

Authoring interaction	How instructors used it in practice	Importance for tangible authoring
Physical element mapping	Assign or redefine meanings for tokens, pages, or modules	Allows instructors to evolve semantics across lessons without rebuilding hardware
Spatial layout authoring	Arrange content and interactions across physical pages and regions	Enables instructors to structure lessons using spatial organization rather than linear flows
Interaction–behavior linking	Map physical actions (placement, touch, gesture) to system responses	Supports embodied explanations where meaning emerges through physical action
Reusable lesson scaffolds	Start from example lessons or templates and adapt them	Reduces blank-slate problem specific to tangible lesson design
Reversible configuration	Reconfigure physical elements during authoring without permanent commitment	Encourages experimentation with tangible interactions

TABLE II: Summary of key tangible authoring practices observed in our studies, including physical element mapping, spatial layout authoring, interaction–behavior linking, reusable lesson scaffolds, and reversible configuration.

VI. DISCUSSION

We designed and implemented TangiTeach, a proof-of-concept authoring environment that enables instructors to create electronics-based tangible lessons without electronics design or wiring. Across two studies with 24 participants, we found that an instructor-centered authoring system can support key tangible authoring capabilities, including structuring content in physical space, reusing components, and iterating on lesson designs. These capabilities align with constructivist perspectives on learning through external representations and scaffolding [20], [21], while remaining grounded in practical system design. By combining wizard-style interfaces with reusable hardware modules, TangiTeach helps instructors focus on lesson design rather than technical setup. Because we did not evaluate student learning outcomes, participants’ comments about engagement and learning should be interpreted as perceptions and opportunities for future classroom use.

A. An Instructor-Centered Approach to Authoring Lessons

Instructors are key to bringing tangible technologies into classrooms [14], [15], yet many face design complexity, time constraints, and limited technical skills [26], [80], particularly in computing education [81], [82]. We extend prior work on tangible authoring by centering instructors’ perspectives and grounding TangiTeach in HCI principles. By lowering technical barriers, TangiTeach enables instructors without electronics experience to create tangible learning resources and transition toward hands-on pedagogy.

Our studies show tools like TangiTeach supporting customizable spatial layouts, embodied input, distributed representations, and shared collaboration allow instructors to focus on pedagogy rather than implementation. From a constructivist view [20], [21], this shift enables instructors to externalize and refine lesson ideas through tangible representations, using space, artifacts, and interaction as scaffolds during the authoring process rather than treating lessons as fixed digital

documents. Future work can explore instructor–student co-creation and identify best practices for both groups.

TangiTeach, as a proof of concept, enabled participants with varied experience, including novices, to successfully create functional tangible lessons. This suggests that instructor-facing tools may encourage adoption among educators willing to experiment, take risks, and share resources [83]. Future directions include a) refining web-based authoring with reusable kits, b) leveraging paper as a low-cost interface, and c) exploring notebook or wizard metaphors for guidance and customization. Together, these directions support more accessible, scalable, and instructor-driven tangible learning environments.

B. Expanding the Scope and Feasibility of TangiTeach

Scaling tangibles to large classrooms has long been a challenge [26], [77], and participants echoed this concern. Yet, they highlighted TangiTeach’s potential as a flexible entry point for integrating tangibles into teaching and envisioned diverse uses: lightweight demos to augment lectures, take-home kits for self-paced exploration, or office-hour companions where students practice with printed booklets that receive cloud-updated feedback. These scenarios suggest feasibility extends beyond classroom scale to hybrid and distributed learning contexts, key concerns for contemporary HCI.

Looking ahead, TangiTeach opens opportunities to broaden the design space of tangible interactions. Technical extensions such as multi-RFID support could enable playful, game-like activities where tokens trigger dynamic events or interact with one another. Supporting simultaneous actions, e.g., placing a token while pressing a button, could further enrich embodied interaction and align with HCI work on multimodal learning. Participants also saw promise in our early explorations of the *Tangible Lens* (Figure 4C) and *gesture-based input* (Figure 7C and Figure 2B), pointing toward kinesthetic techniques that could deepen engagement and diversify pedagogical strategies.

Together, these directions suggest that the feasibility of tangible learning is not only a matter of classroom logistics, but also of designing interaction techniques that balance scalability with meaningful, embodied experiences.

C. Designing Inspiration for Lesson Creation and Sharing

A recurring theme across both studies was that instructors rarely struggled with TangiTeach’s interface or technical setup, but with the blank-slate challenge of authoring lesson content. For many, the time and creativity needed to design meaningful lessons could hinder adoption, though participants were enthusiastic about two complementary sources of inspiration: examples from peers and built-in AI-powered suggestions.

A community lesson bank with shared examples, ratings, and institutional context was seen as reducing design overhead and sparking ideas. This aligns with research on communities of practice in teaching [69], [70], [84], where collaboration supports efficiency and innovation. In computing education, prior work highlights a culture of community-driven exchange [85], [86]. TangiTeach can facilitate collaboration and remixing, while enabling students to contribute lessons, enriching

the repository and reducing the need for instructors to design content from scratch.

Although TangiTeach includes AI features, they primarily serve as authoring scaffolds that complement instructors’ workflows. Participants viewed AI as a creative partner for generating examples and personalized content, consistent with prior work on inquiry-based lesson planning and programming pedagogy [73], [74], [79]. While overreliance remains a concern [87], our findings suggest AI supports inspiration rather than replacing pedagogy. Future work could combine AI-generated lesson ideas with recent advances in editable 2D graphics [88], [89] to produce printable manipulatives.

VII. LIMITATIONS

We created TangiTeach as a proof-of-concept to support instructors in authoring electronics-based interactive tangibles. Our studies examined a limited set of lessons and modules, although the platform is designed to support additional interactions and content. The use of letter-size paper constrains the interaction space and number of detectable signals, while the emphasis on simplicity and low cost limits display space and support for more complex interaction logic. Although our qualitative studies provided rich insights into instructors’ needs and authoring practices, future work should include controlled studies with larger and more diverse educator populations across subjects beyond programming. Our evaluation demonstrated collaborative lesson creation using instructor–student pairs, but future work should also investigate larger collaborative settings, classroom adoption, lesson quality, sustained use, and recognition robustness under real-world conditions such as varying lighting, page positioning, and repeated handling. Future versions could incorporate richer sensing, displays, and interaction techniques while maintaining ease of use.

VIII. CONCLUSIONS

This paper introduced TangiTeach, an end-to-end authoring tool that enables instructors without coding or electronics expertise to create tangible lessons. Across two studies, instructors successfully explored interactive lessons, demonstrating the feasibility of instructor-centered tangible authoring. TangiTeach combines reusable modules, a paper-based metaphor, wizards, and inspiration scaffolds to support key authoring capabilities, including structuring content in physical space, reusing components, and iterating on lesson ideas. As a proof of concept, our findings provide empirical insights into instructor practices and challenges, informing future tangible authoring systems and flexible physical interaction workflows.

IX. ACKNOWLEDGMENTS

We thank the Natural Sciences and Engineering Research Council of Canada (NSERC).

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TangiTeach was designed using inexpensive, off-the-shelf components rather than a custom PCB. The current prototype costs approximately USD \$25, including a Lolin Wemos D32 Lite module (\$3.32), a 2.42-inch OLED display (\$11.02), an RC522 RFID reader with 10 RFID stickers (\$4.01), an Arduino Nano (\$1.14), and TTP223 touch sensors (\$4.79 for 20 pieces). The prototype requires approximately 2–3 hours to assemble. Across our studies, page and RFID recognition achieved approximately 95% accuracy, supporting successful lesson creation and testing.

A. Modular and Expandable Design

A key advantage of TangiTeach is its modular design. Each physical module is a flat, slim self-contained rectangle, with its own controller that can be swapped or arranged in any of the four available spots in the platform (Figure 7A). Using the standard I2C protocol with an ATmega328P microcontroller as an interface with the main platform (Figure 7H), TangiTeach supports expansion with new modules for different interactions. In our first iteration, we developed three different removable modules:

- **Multiple Choice Question Module:** This module uses a TTP223 capacitive sensor (Figure 7G) to detect touches on any of five spots. Users can select by touching the paper, and the platform will display feedback for that choice on the OLED (Figure 7D) screen, feedback that has been provided by the web authoring tool.
- **Matching Question Module:** Similar to the multiple-choice module, this has 10 spots in two columns. The user touches one spot in each column to create a match, and the platform displays feedback for the selected combination on the OLED screen.
- **Object Placement Module:** This module enables a kinesthetic experience by detecting objects like cards (Figure 4A) or cubes (Figure 4B) with RFID stickers, using an RC522 RFID module (Figure 7G). Each object triggers

actions based on its RFID tag, and the module supports up to 20 stickers per page.

Additionally, we include two modules to provide enhanced kinesthetic interactivity:

- **Hand Gestures:** Our platform is capable of detecting hand gestures using two fixed VL53L0X distance sensors near the top left and right corners of the platform (Figure 7C), allowing to detect from 5cm to 100cm of distance. By using both sensors in a sequence, and through our authoring interface (Figure 2B), users can assign actions to hand gestures (e.g. swiping left to right, right to left, and making circles over the sensors). There are options to trigger specific messages based on the frequency of each gesture. For example, a student's circular hand motion can simulate adding an extra orange to the total weight on a virtual scale. The screen then shows the weight increasing until it reaches a desired amount, demonstrating how a loop functions.
- **Tangible Lens Device:** To extend interactions, we developed a “Tangible Lens” (Figure 4C), a cardboard device with a 2.4” OLED screen, ESP32, 1000mhz battery, and RC522 RFID reader. It reads RFID stickers strategically places on platform modules and, via Wi-Fi, displays configurable messages defined in the authoring interface. The lens offers supplementary explanations—for example, in an object-oriented programming lesson, students can place tokens to reveal examples of different methods.

B. Usability Tasks

Our main goal in designing the TangiTeach authoring tool was to enable the creation of end-to-end content for an electronics-based interactive tangible programming lesson without having to directly program the underlying electronics. To evaluate this, we conducted our study with three key tasks: 1) a warm-up task to understand how TangiTeach works from the students' perspective; 2) a task to familiarize participants with the web authoring UI; and, 3) our main task where participants created their own tangible lesson



Fig. 6: This figure shows how G8 interacted with their created lesson. A) Selecting a multiple-choice answer triggers immediate on-screen feedback: “Right” appeared on the screen. Notice the stack of Tangible Tokens on the right side; these were: *Code Reusability*, *Modularity*, *Readability & Maintainability*, and *Abstraction*; B-C) Tangible tokens placed on the hardware surface reveal definitions and explanations (e.g., for *Code reusability*, the screen shows: “You don’t have to repeat code, you can keep using the same function.” Participants valued the immediacy and reported this as their first experience building a tangible lesson.

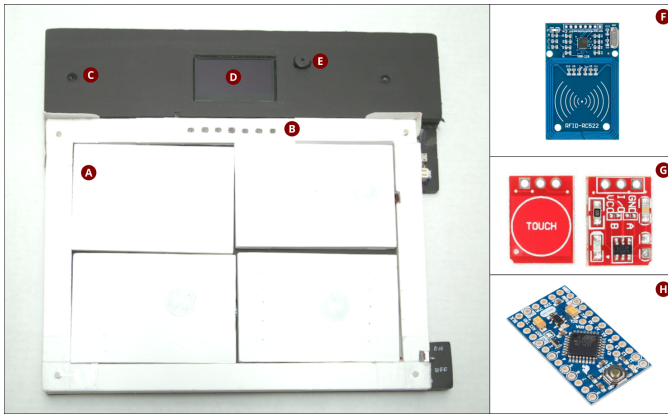


Fig. 7: TangiTeach Hardware Platform features a low-cost, modular design for reuse and reconfiguration. A) Four modules (Cover, Object Placement, Multiple Choice, Matching) built with off-the-shelf electronics and foam-board, supporting printed overlays. B) Page Detection: Seven light sensors identify pages and auto-load lessons. C) Hand Gestures: Two sensors per side enable gesture input. D) OLED Screen displays feedback. E) Buzzer provides audio feedback. F) RC522 RFID under the object placement module. G) TP23 modules enable touch. H) ATmega328 per module supports I2C for flexible placement and expansion.

using TangiTeach. These tasks were designed based on three pilot studies conducted before the main study to ensure an appropriate balance of challenge, time, novelty, and feature exploration. The goal was to gather valuable insights into users’ mental models, reactions, and perceptions.

Task 1: Interacting with a TangiTeach-based lesson:

The objective of this task was to familiarize participants with the TangiTeach platform and to browse a pre-made lesson. During this task (5 minutes max), participants explored the capabilities of the TangiTeach hardware, learned about its different components, and interacted with its hardware modules and tangible tokens. Participants were given a set of printed pages with a lesson about loops in everyday situations, which they loaded onto the platform on their own. The first page contained definitions and examples, and the second page was about different types of loops. On this page, participants had different physical tokens that represented the main loops: for, while, for-each, and do-while (Figure 4B). The student participant led this task.

Task 2: Understanding lesson components and getting familiar with the TangiTeach web-based UI: This task required participants to use the TangiTeach web-based authoring tool to modify an existing lesson about loops and self-testing questions. The task (5 minutes max) asked participants to change one of the messages that appears when the student selects an incorrect option in a self-testing question. Finally, the TangiTeach hardware platform was used to demonstrate how the changes made in the UI automatically appear on the hardware platform, thus reflecting how input actions lead to learning outputs. The instructor led this task.

Task 3: Creating end-to-end tangible lessons using the TangiTeach platform: The main task (30 minutes max) asked participants to create a suitable lesson about *functions* in programming. From that request, they were free to add content

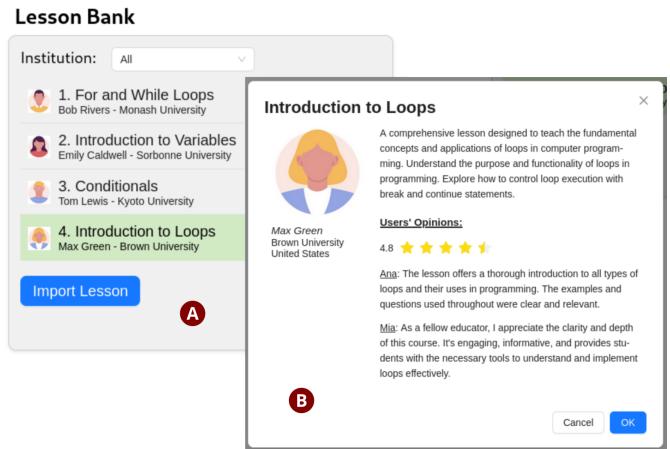


Fig. 8: Community Lesson Bank: A) Instructors can browse a lesson bank with lesson names, authors, and institution filters. B) Users can view lesson details: author description, ratings, and comments. This feature reduces blank-slate effort and supports reuse of trusted resources, directly addressing instructors’ time barriers.

such as definitions, examples, activities with different degrees of lesson difficulty. Participants were able to browse lessons from a lesson bank and filter by different criteria such as institutions or authors. We let them explore the community-based features (e.g., rankings, reviews, and descriptions) to assess the lesson before importing the content into their own lesson. Participants were asked to add at least one new page in their lesson to enhance the topic of functions in programming and to add at least two self-testing questions, including using a new set of tangible tokens if they wished.

For creating this lesson and its content, participants had a range of options such as a) searching Google for examples, b) using our tool’s built-in AI features, c) importing questions from our question bank, or d) using the printed material we provided for inspiration. This material consisted of some textbook content titled *Why use functions in programming?*, with a real-life example of a function, `juiceInOrange(circumference)`, to calculate the amount of juice that any given orange can provide, and a short list of the benefits of using functions in programming. We also provided an optional list of *“Possible ideas for self-testing questions”* with more question ideas. We provided this content for inspiration because we were not testing our participants’ ability to create learning resources, but rather their perceptions of the overall authoring process with TangiTeach. The instructor led this task.